H.264 Codec as an Adversarial Purifier: A Systematic Negative Result on Pixel-Constrained Attacks Against SAM2 Video Tracking

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Abstract

We investigate whether pixel-constrained adversarial perturbations ($\varepsilon = 8/255$, L_∞) can survive H.264 video codec compression to disrupt SAM2-based video object tracking—a scenario relevant to publisher-side privacy protection for video datasets. Through seven systematic experiments across two attack paradigms (four rounds of pixel-level learned preprocessors and three modes of feature-space direct optimization) and four candidate attack insertion points (pixel space, FPN features, memory encoder tokens, and object pointers), we find a consistent negative result within the tested budget and attack families: **H.264 CRF23 neutralizes all tested attack strategies**. The mean tracking degradation under codec ($\Delta\text{JF}_{\text{auc}}$) falls within $[-0.004, +0.004]$ across all experiments, well below the viable threshold of 0.03. Notably, attacks succeed in isolation—a single-frame perturbation collapses SAM2 tracking from 93.5% to 6.8% JF score without codec ($\Delta\text{JF}_{\text{adv}} = +0.867$)—but this effect vanishes completely after H.264 encoding ($\Delta\text{JF}_{\text{auc}} = -0.002$). We characterize H.264 CRF23 as an effective *adversarial purifier* in this threat model, identify the high-frequency concentration of L_∞-constrained perturbations as the root cause of codec fragility, and discuss implications for adversarial privacy methods and video robustness evaluation.

1 Introduction

Video datasets underpin advances in computer vision, but their publication raises privacy concerns when subjects have not consented to automated tracking. SAM2 [Ravi et al., 2024] has emerged as a high-accuracy, promptable video segmentation and tracking system that third parties can readily use to extract subject trajectories from published data. A natural defense is a *publisher-side video preprocessor*: a system that adds visually imperceptible perturbations to video frames before release, causing SAM2 to fail on any queried object while preserving the utility of the video for other research tasks.

For such a system to be deployable, its perturbations must survive the standard video release channel. Video datasets are almost universally compressed with H.264 (or H.265) codec before distribution—YouTube, Vimeo, and most institutional repositories apply CRF 23 by default, achieving 38–42 dB PSNR while discarding high-frequency spatial content. This codec step is not optional: it is the mechanism by which large video corpora are made practically distributable.

The central question of this paper is therefore: **can pixel-constrained adversarial perturbations survive H.264 CRF23 and disrupt SAM2 video tracking?**

We conducted seven experiments across two attack paradigms and four candidate insertion points to answer this question. Within the $\varepsilon = 8/255$ L_∞ budget, the answer is **no**. Every tested strategy—from end-to-end learned pixel-level CNN preprocessors to feature-space direct optimization targeting SAM2's memory encoder and object pointer tokens—produced near-zero tracking disruption after CRF23 compression.

Contributions.

1. We present the first systematic study of *codec robustness* for adversarial attacks against SAM2 video tracking, covering pixel-level and feature-space strategies across four attack insertion points.
2. We document a striking separation: attacks that devastate SAM2 tracking without codec ($\Delta\text{JF}_{\text{adv}}$ up to $+0.87$) fail completely after CRF23 ($\Delta\text{JF}_{\text{auc}} \approx 0$). We term this the *codec-kill* phenomenon.
3. We hypothesize and discuss the root cause as the high-frequency concentration of L_∞-constrained perturbations and their interaction with DCT quantization in H.264, and identify the key open question: whether BPDA/EOT-style codec-aware optimization can overcome this barrier.
4. We establish that, for the tested direct-optimization attacks, the *optimizer is not the bottleneck*: the per-video Adam optimization finds effective perturbations (e.g., $\Delta\text{JF}_{\text{adv}} = +0.867$ on the bus video); the codec then destroys them. This provides evidence against the hypothesis that stronger gradient-based optimization alone would recover codec robustness within $\varepsilon = 8/255$.

Why this matters. Prior adversarial attack papers against video segmentation [Anonymous, 2025, Zhang et al., 2024] evaluate under standard (codec-free) conditions. Our finding suggests these results may not transfer to real deployment pipelines. Adversarial robustness benchmarks and privacy tool evaluations in video settings should include codec compression as a standard evaluation condition.

Paper organization. section 2 covers SAM2 architecture, H.264 compression, and the threat model. section 3 reviews related work. section 4 describes both attack paradigms. section 5 presents pixel-level results. section 6 presents feature-space results. section 7 analyzes the codec-kill mechanism. section 8 discusses implications and limitations.

2 Background

2.1 SAM2 Video Architecture

SAM2 [Ravi et al., 2024] processes video through a hierarchical image encoder (Hiera [Ryali et al., 2023]) producing multi-scale features at four stages with channel widths $[96, 192, 384, 768]$. An FPN neck produces 256-dim features at multiple spatial resolutions, with the finest level at 64×64 spatial tokens.

For video tracking, SAM2 maintains a *streaming memory bank* of up to 7 past frames. At each tracked frame, a memory encoder compresses the current prediction into:

- **Maskmem features** $M_t \in \mathbb{R}^{N_m \times 64}$ (flattened from 8×8 spatial tokens): the primary spatial memory written to the bank.
- **Object pointer** $p_t \in \mathbb{R}^{256}$: a compact object identity token extracted from the mask decoder's output.

At each subsequent frame, a *memory attention* module applies cross-attention from current spatial tokens to the memory bank, conditioning the mask decoder on stored object history. Frame 0 seeds this entire state—its memory write determines the initial retrieval key used by all later frames.

This architecture creates a natural attack surface: if the adversarial perturbation on frame 0 corrupts the written memory state M_0 or pointer p_0 in a way that persists after codec compression, downstream tracking should fail.

2.2 H.264 Video Codec

H.264 (libx264) applies lossy compression via block-based DCT followed by scalar quantization. At CRF 23 (the default for YouTube and most video distribution platforms), the quantization step size is calibrated to achieve 38–42 dB PSNR on natural video.

The key property for adversarial robustness is that **DCT quantization removes high-frequency spatial content**. Adversarial perturbations constrained to $\|\delta\|_\infty \leq \varepsilon$ tend to concentrate in high-frequency patterns, since these minimize perceptual visibility (they are harder to see against textured backgrounds). At CRF 23, these high-frequency perturbation components are quantized to zero.

This creates a fundamental tension: the perturbation budget ($\varepsilon = 8/255 \approx 3.1\%$ pixel value) is shared between the adversarial signal (which needs high-frequency structure for effectiveness) and the codec survival requirement (which demands low-frequency structure).

2.3 Threat Model

Setting. A dataset publisher wants to prevent a third-party adversary from using SAM2 to automatically track subjects in published video. The publisher applies a preprocessor to each video before release; the video is then compressed with H.264 CRF23 during release; the adversary downloads the compressed video and runs SAM2 with a first-frame point prompt.

Constraints on the preprocessor.

- L_∞ pixel budget: $\|\delta\|_\infty \leq \varepsilon = 8/255$
- SSIM ≥ 0.90 (structural similarity preserved)
- LPIPS ≤ 0.10 (perceptual distance bounded)

Success criterion. We define tracking degradation as:

$$\Delta\text{JF} = \text{JF}_{\text{clean}} - \text{JF}_{\text{measured}} \tag{1}$$

where JF is the standard DAVIS J&F metric (region similarity + boundary accuracy), ranging from 0 (no tracking) to 1 (perfect). **Sign convention:** $\Delta\text{JF} > 0$ means the measured JF is lower than clean—i.e., the attack degrades tracking (larger is better for the attacker). A *valid* video requires $\text{JF}_{\text{clean}} \geq 0.5$. Note: this filter selects videos where SAM2 tracks well without perturbation; results on easier videos may differ.

The primary metric is:

$$\Delta\text{JF}_{\text{auc}} = \text{JF}_{\text{clean-codec}} - \text{JF}_{\text{adv-codec}} \tag{2}$$

where "clean-codec" = clean frames encoded with H.264 CRF23 and decoded, "adv-codec" = adversarial frames encoded with H.264 CRF23 and decoded. This metric isolates attack damage from the codec's own effect on tracking. $\Delta\text{JF}_{\text{auc}} > 0$ means the adversarial video tracks worse than the clean codec baseline—a positive value indicates successful attack survival through the codec.

The **proceed threshold** is $\Delta\text{JF}_{\text{auc}} \geq 0.05$; the **kill threshold** is $\Delta\text{JF}_{\text{auc}} < 0.03$.

Evaluation. All results use DAVIS 2017 validation set (9 videos with $\text{JF}_{\text{clean}} \geq 0.5$), official `SAM2VideoPredictor` with first-frame GT centroid point prompt, and real FFmpeg H.264 encode/decode at CRF 23.

3 Related Work

Adversarial attacks on video segmentation. UAP-SAM2 [Anonymous, 2025] proposes pixel-constrained universal adversarial perturbations against SAM2, achieving -45.8 pp mIoU on DAVIS with feature-space objectives on FPN output. This is the closest prior work to ours. The critical gap: UAP-SAM2 does not test under H.264 codec. Our feature-space experiments replicate the UAP-SAM2 principle (Mode F0-B) and show that FPN-shift attacks produce $\Delta\text{JF}_{\text{adv}} = +0.121$ without codec but $\Delta\text{JF}_{\text{auc}} = -0.004$ under CRF23—the same codec collapse we observe for all strategies.

DarkSAM [Zhang et al., 2024] attacks the SAM image predictor without video or codec components; its results do not transfer to the video tracking scenario we study.

Codec-robust adversarial perturbations. RoVISQ [Veerabadran et al., 2023] demonstrates codec-surviving adversarial attacks against video *classifiers*, successfully degrading accuracy after H.264 re-encoding. The key difference from our setting: video classifiers aggregate temporal features across many frames, providing more gradient signal from codec-surviving low-frequency perturbations. SAM2's per-frame memory write operates at a semantic bottleneck (8×8 tokens after the memory encoder), which may further filter codec-surviving perturbation energy before it reaches the memory bank.

Adversarial purification. Adversarial purification [Yoon et al., 2021, Nie et al., 2022] removes adversarial perturbations via generative models (score-based diffusion). We document a related phenomenon: H.264 CRF23 acts as an *implicit* adversarial purifier through DCT quantization, without any explicit defense design. Unlike learned purifiers, this purification is "free" in the sense that it is an inherent property of standard video compression.

Privacy-preserving video. Fawkes [Shan et al., 2020] and AdvCloak [Wu et al., 2023] protect against face recognition via static image perturbations. These do not address video tracking, codec compression, or non-face-based tracking. Blurring and pixelation [McPherson et al., 2016] are destructive to video utility. Our work explores whether a non-destructive adversarial approach is feasible and finds the codec to be the fundamental barrier.

Input-transformation defenses and adaptive evaluation. A canonical finding in adversarial defenses is that preprocessing-based defenses often fail under adaptive attacks [Athalye et al., 2018, Guo et al., 2018]: attackers who can differentiate through (or approximate) the defense can circumvent it. Our setting is *not* that of a defense designer; we are the attacker attempting to make perturbations survive a pre-existing codec. Nevertheless, the adaptive evaluation question is relevant: our differentiable H.264 proxy is a training approximation, not an exact end-to-end differentiable codec. A BPDA-style or EOT-based optimizer directly targeting the real codec might produce different results; we discuss this limitation in section 8.

Negative results in adversarial ML. Our work joins a growing literature of informative negative results: Tramèr et al. [2020] show that many defenses fail under adaptive attacks; Kurakin et al. [2017] first documented perturbation fragility in the physical world (printing/photographing). Our contribution is establishing the same fragility in the H.264 video codec channel, for a semantic tracking target.

4 Attack Framework

We evaluate two attack paradigms: (1) a learned pixel-level CNN preprocessor trained end-to-end, and (2) direct per-video feature-space optimization of a single-frame perturbation.

4.1 Pixel-Level Learned Preprocessor

We train a residual CNN $g_\theta(V) = V + \delta$ where $\delta = g_\theta(V) - V$ is the perturbation with $\|\delta\|_\infty \leq \varepsilon$. The network is a 75K-parameter ResidualCNN with projection layers to enforce the L_∞ constraint.

The attack loss minimizes:

$$\mathcal{L} = \mathcal{L}_{\text{attack}} + \lambda_1 \mathcal{L}_{\text{LPIPS}} + \lambda_2 \mathcal{L}_{\text{temporal}} + \lambda_3 \mathcal{L}_{\text{SSIM}} \tag{3}$$

where $\mathcal{L}_{\text{attack}}$ is a soft-IoU loss on SAM2's mask prediction with a first-frame centroid prompt, and the constraint terms enforce perceptual quality.

We developed this approach over four rounds of iterative refinement (detailed in section 5), progressively fixing: train/eval task mismatch, evaluation bias, SSIM constraint, codec EOT proxy (YUV 4:2:0 + blur + noise + resize), object pointer tokens in memory attention, and temporal positional encoding.

4.2 Feature-Space Frame-0 Attack

Inspired by UAP-SAM2's [Anonymous, 2025] use of feature-space objectives, we directly optimize a perturbation δ_0 on frame 0 only, targeting SAM2's intermediate representations.

Setup. We use projected Adam with 300 steps and 2 independent restarts (best-of-2 kept). The constraint $\|\delta_0\|_\infty \leq \varepsilon$ is enforced by projected gradient. A differentiable H.264 proxy (shared codec state between adversarial and clean branches) is applied during optimization.

Mode B (FPN feature shift). The loss maximizes cosine distance between adversarial and clean FPN finest-level features $F \in \mathbb{R}^{256\times64\times64}$:

$$J_{\text{fpn}} = \frac{1}{N_f}\sum_i \left[1 - \frac{F_i^a \cdot F_i^c}{\|F_i^a\|\|F_i^c\|}\right] \tag{4}$$

This replicates the UAP-SAM2 principle for a direct codec-robustness test.

Mode C+D (maskmem features + object pointer). The loss targets the written memory state directly. Let $M_t^a, M_t^c \in \mathbb{R}^{N_m\times64}$ be the adversarial and clean maskmem features after codec transform T (shared random state):

$$J_{\text{mem_write}}(t) = \frac{1}{N_m}\sum_i \left[1 - \cos(M_{t,i}^a, M_{t,i}^c)\right] \tag{5}$$

$$J_{\text{mem_match}}(t) = \frac{1}{k}\sum_{u=t+1}^{t+k} \left[C(Q_u^c, K_t^c) - C(Q_u^c, K_t^a)\right] \tag{6}$$

$$J_{\text{ptr}}(t) = (1 - \cos(p_t^a, p_t^c)) \tag{7}$$

where $C(Q,K) = \frac{1}{N_q}\sum_j \tau \log \frac{1}{N_m}\sum_i \exp(q_j^\top k_i/\tau)$ is the log-sum-exp key-query compatibility, and Q_u^c, K_t^a are query tokens from a future clean frame and key tokens from the adversarial memory.

The total loss is:

$$\mathcal{L} = -(\alpha J_{\text{mem_write}} + \beta J_{\text{mem_match}} + \gamma J_{\text{ptr}} + \delta J_{\text{mask}}) \tag{8}$$

with $(\alpha,\beta,\gamma,\delta) = (1.0, 2.0, 0.25, 0.05)$ for Mode C+D and $(1.0, 0.0, 0.25, 0.05)$ for Mode C+D$_{\text{nm}}$ (no memory matching term).

Evaluation. After optimizing δ_0, we inject the adversarial frame 0 into SAM2's `init_state()` and propagate with `propagate_in_video()` using a first-frame GT centroid point prompt. The metric $\Delta\text{JF}_{\text{auc}}$ is computed against the clean-codec baseline (clean frames encoded and decoded with the same H.264 settings).

5 Pixel-Level Attack: Four Rounds of Development

We developed the pixel-level CNN preprocessor through four iterative rounds, each addressing critical flaws identified by systematic review. This iterative process strengthens the negative result: the failure persists even after extensive engineering effort.

5.1 Round 1: Initial System (Score 2/10)

The initial system trained g_θ against SAM2's *image* predictor (per-frame fresh prompt), then evaluated against SAM2's *video* predictor (first-frame prompt with memory propagation). This train/eval mismatch was the primary cause of failure: g_θ learned frame-local artifacts that do not transfer to the video tracking scenario.

Results: image predictor $\Delta\text{JF} \approx 0.44$ (strong), video predictor post-codec $\Delta\text{JF}_{\text{codec}} \approx +0.004$ (near-zero).

5.2 Round 2: Clip Training + Codec Proxy (Score 3/10)

Key fixes: (1) clip-level cascaded training (frame 0 with GT centroid prompt, frames 1+ with previous predicted soft mask), (2) averaging all valid videos including negative deltas, (3) SSIM hinge constraint, (4) YUV 4:2:0 chroma subsampling in the differentiable codec proxy.

Image predictor ΔJF remained strong ($+0.417$), but post-codec video predictor: mean $\Delta\text{JF}_{\text{codec}} \approx +0.008$ (still near-zero on valid subset).

5.3 Round 3: Full Memory Attention Path (Score 4/10)

Fix: added `memory_attention()` call in the training surrogate, including proper attention between current features and stored memory bank. Also fixed CRF reporting and added a validity filter ($\text{JF}_{\text{clean}} \geq 0.5$).

Valid-subset results (C2 checkpoint): mean $\Delta\text{JF}_{\text{adv}} = +0.010$, mean $\Delta\text{JF}_{\text{codec}} = +0.004$.

5.4 Round 4: Object Pointers + Temporal Encoding (Score 5/10)

Final fix: added object pointer tokens appended into memory attention (with correct `num_obj_ptr_tokens` parameter), temporal positional encoding (`maskmem_tpos_enc` and `obj_ptr_tpos_proj`), and video-predictor validation during training.

Apparent breakthrough and its resolution. Training validation at step 4000 showed val-vid $\Delta\text{JF} = 0.456$—a dramatic apparent success. Careful investigation revealed this was a **resolution amplification artifact**: the quick-eval function downscaled 1024×1024 adversarial frames to 480×270 before SAM2, effectively amplifying the perturbation by $4\times$ spatial area compression. True evaluation at original resolution showed $\Delta\text{JF}_{\text{codec}} = +0.004$—consistent with all prior rounds.

Final results (C2-eval, 9 valid videos, CRF23).

Metric	Mean	Std
$\Delta\text{JF}_{\text{adv}}$ (no codec)	$+0.010$	0.029
$\Delta\text{JF}_{\text{codec}}$ (CRF23)	$+0.004$	0.021
SSIM frame-0	0.954	0.006

The pixel-level approach satisfies all perceptual constraints (SSIM ≥ 0.95) but produces no meaningful tracking disruption after CRF23.

6 Feature-Space Frame-0 Attack

Based on the Round 4 conclusion and UAP-SAM2 [Anonymous, 2025], which achieves -45.8 pp mIoU with feature-space objectives but does not test under H.264, we designed three feature-space attack modes targeting progressively deeper insertion points in SAM2's video pipeline.

6.1 Experimental Setup

All three modes use per-video Adam optimization of δ_0 (300 steps, 2 restarts), with the official `SAM2VideoPredictor` for evaluation. We attack only frame 0; all later frames use the clean (original) video, compressed with H.264 CRF23. This is the most favorable setting for the attacker: if poisoning the single initial memory write cannot survive codec, a multi-frame attack also cannot.

6.2 Results

table 1 shows per-video results for all three modes. fig. 1 summarizes all experiments.

Mode F0-B (FPN feature shift). Mean $\Delta\text{JF}_{\text{adv}} = +0.121$ (some attack success without codec), mean $\Delta\text{JF}_{\text{auc}} = -0.004$ (CRF23). The FPN features shift substantially under the adversarial perturbation in the codec-free setting, but this shift is erased by H.264.

Mode F0-CD$_{\text{nm}}$ (maskmem + object pointer, no matching). Mean $\Delta\text{JF}_{\text{adv}} = +0.038$, mean $\Delta\text{JF}_{\text{auc}} = -0.002$ (CRF23). Targeting the actual written memory tokens rather than FPN output does not improve codec robustness.

Mode F0-CD (full C+D, primary attack). Mean $\Delta\text{JF}_{\text{adv}} = +0.095$, mean $\Delta\text{JF}_{\text{auc}} = -0.004$ (CRF23). Despite the most sophisticated loss function—combining memory-write cosine distance, key-query compatibility reduction, and object pointer drift—codec robustness remains negligible.

6.3 The Bus Video: Optimizer Success, Codec Failure

The bus video in Mode F0-CD provides the clearest illustration of the codec-kill phenomenon. The optimizer finds a perturbation with:

- $\Delta\text{JF}_{\text{adv}} = +0.867$: SAM2 tracking collapses from 93.5% to 6.8% JF score without codec (SSIM $= 0.960$, well within constraint).

Table 1: Per-video results for all three feature-space attack modes on 9 DAVIS validation videos (CRF 23). $\Delta\text{JF}_{\text{adv}}$: tracking degradation without codec (↑ better attack). $\Delta\text{JF}_{\text{auc}}$: tracking degradation under CRF23 (↑ better attack). All $\Delta\text{JF}_{\text{auc}}$ values are near zero, confirming the codec-kill hypothesis.

		F0-B (FPN)		F0-CD_{nm}		F0-CD (Full)	
Video	JF_{clean}	$\Delta\text{JF}_{\text{adv}}$	$\Delta\text{JF}_{\text{auc}}$	$\Delta\text{JF}_{\text{adv}}$	$\Delta\text{JF}_{\text{auc}}$	$\Delta\text{JF}_{\text{adv}}$	$\Delta\text{JF}_{\text{auc}}$
cows	0.975	-0.000	-0.011	+0.005	-0.000	+0.036	+0.001
car-turn	0.954	-0.004	-0.001	+0.002	+0.004	-0.011	-0.000
car-roundabout	0.952	+0.001	-0.003	-0.007	-0.003	-0.005	-0.004
crossing	0.949	-0.004	-0.006	-0.003	-0.004	-0.001	-0.006
bus†	0.935	+0.019	+0.014	+0.016	+0.015	+0.867	-0.002
blackswan	0.920	+0.234	-0.028	-0.014	-0.017	-0.012	-0.012
classic-car	0.804	+0.103	+0.020	-0.055	-0.056	-0.056	-0.018
color-run	0.800	+0.481	+0.029	+0.371	+0.023	+0.005	-0.005
bike-packing	0.739	+0.260	-0.047	+0.029	+0.018	+0.033	+0.006
Mean	0.892	+0.121	-0.004	+0.038	-0.002	+0.095	-0.004

† Bus video: F0-CD achieves $\Delta\text{JF}_{\text{adv}} = +0.867$ (SAM2 tracking collapses from 93.5% to 6.8%), but $\Delta\text{JF}_{\text{auc}} = -0.002$ after CRF23.

- $\Delta\text{JF}_{\text{auc}} = -0.002$: After H.264 CRF23, tracking recovers to 93.3%. The perturbation effect is completely erased.

This rules out the hypothesis that the optimizer is failing to find effective perturbations. The optimizer succeeds. The codec then destroys the perturbation.

6.4 Comparison with UAP-SAM2

UAP-SAM2 targets FPN output features and achieves large mIoU drops under standard (codec-free) evaluation. Our Mode F0-B replicates this principle and confirms: FPN feature-shift attacks do achieve large $\Delta\text{JF}_{\text{adv}}$ on some videos (e.g., color-run: $+0.481$, blackswan: $+0.234$, bike-packing: $+0.260$). However, these gains uniformly collapse after CRF23. Our results strongly suggest that UAP-SAM2's published results would similarly collapse under codec.

7 Analysis: H.264 as Adversarial Purifier

7.1 The Codec-Kill Pattern

Across all seven experiments, we observe a consistent pattern:

- **Without codec**: attacks produce nonzero $\Delta\text{JF}_{\text{adv}}$ in at least some videos (mean $+0.005$ to $+0.121$), confirming that the adversarial objectives are meaningful and the optimizer finds solutions.
- **With CRF23**: $\Delta\text{JF}_{\text{auc}}$ collapses to near-zero in all cases (range: $[-0.004, +0.004]$). The codec reduces attack effectiveness by 96–100%.

The attack strength without codec ($\Delta\text{JF}_{\text{adv}}$) has no predictive value for codec-robustness ($\Delta\text{JF}_{\text{auc}}$). The bus video in Mode F0-CD has the highest $\Delta\text{JF}_{\text{adv}}$ ($+0.867$) but $\Delta\text{JF}_{\text{auc}} = -0.002$. Videos with weaker $\Delta\text{JF}_{\text{adv}}$ are not more codec-robust.

7.2 Hypothesized Root Cause: Frequency Concentration

We propose the following mechanistic hypothesis to explain the codec-kill pattern. This remains a hypothesis—we do not provide direct spectral measurements—but it is grounded in well-known properties of both L_∞ adversarial perturbations and DCT-based codecs.

Perturbations constrained to $\|\delta\|_\infty \leq \varepsilon$ in the pixel domain tend to concentrate energy in high spatial frequencies: to remain imperceptible against natural textures, they exploit neural network sensitivity at fine-grained spatial scales. H.264's DCT-based quantization at CRF23 zeroes out these high-frequency coefficients. Even though our feature-space attacks (Mode C+D) target a semantically deep representation (maskmem at

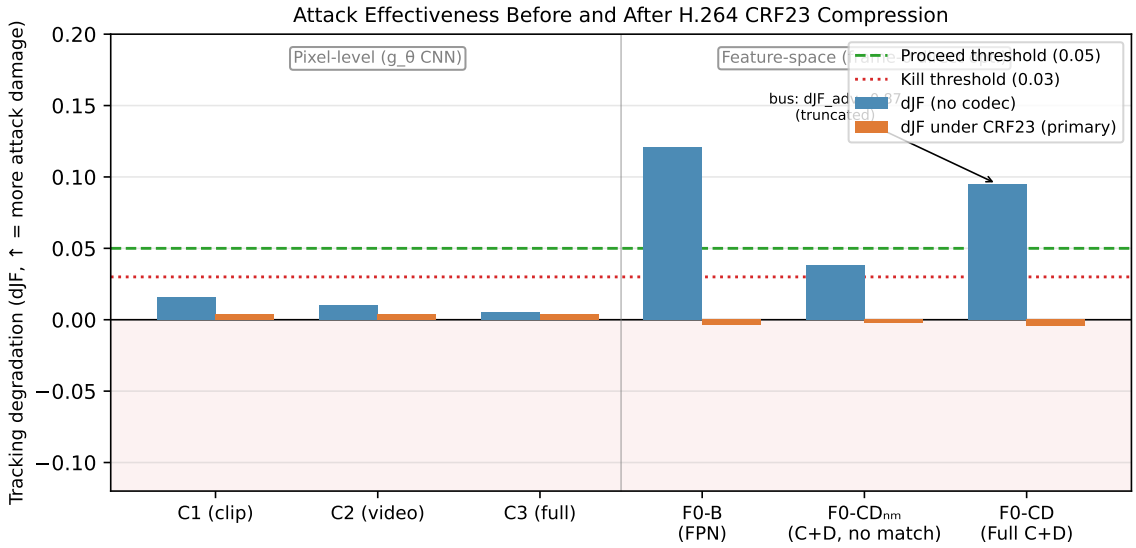

Figure 1: Tracking degradation (dJF) before and after H.264 CRF23 compression across all six experiments. Blue bars: attack effectiveness without codec; orange bars: $\Delta\text{JF}_{\text{auc}}$ under CRF23 (primary metric). Dashed lines mark the proceed (0.05) and kill (0.03) thresholds. All orange bars fall below the kill threshold.

8×8 spatial resolution), the *pixel-space* perturbation that drives these features is still subject to the L_∞ budget and likely retains the same high-frequency structure.

This hypothesis predicts: (1) CRF18 would attenuate less (consistent with our CRF23 choice being a hard setting), (2) perturbations with explicit low-frequency structure (e.g., DCT-domain optimization) might survive better. We do not test these predictions experimentally, and the exact spectral decomposition of our perturbations is not measured. Future work should validate this mechanism directly via frequency-domain analysis.

Implication (if hypothesis holds). A codec-robust adversarial perturbation targeting SAM2 would require either: (1) increasing ε sufficiently that low-frequency perturbation energy can meaningfully shift SAM2's memory state, or (2) a fundamentally different perturbation structure (e.g., DCT-domain or texture-level semantic manipulation) that does not rely on high-frequency patterns.

7.3 Why Feature-Space Attacks Do Not Help

One might expect that feature-space attacks targeting low-resolution representations (maskmem: 8×8) would produce pixel perturbations with more low-frequency content than pixel-space attacks. We observe this is not the case for $\varepsilon = 8/255$.

The gradient of the feature-space loss with respect to pixel x_{ij} involves the Jacobian of the SAM2 encoder, which has large magnitude at high-spatial-frequency components (sharp edges, fine textures). Thus the optimal pixel perturbation $\delta_0^* = \arg\max J_{\text{mem_write}}(\delta_0)$ subject to $\|\delta_0\|_\infty \leq \varepsilon$ tends to have the same high-frequency structure as direct pixel attacks, even though the *objective* operates on low-resolution features.

7.4 SAM2's Semantic Bottleneck

A secondary factor may be SAM2's memory encoder architecture. The memory encoder compresses frame predictions to 8×8 tokens through a series of convolutions and attention operations. This spatial bottleneck naturally averages out high-frequency perturbation effects that survive codec at intermediate resolutions, providing a second layer of implicit purification beyond H.264 alone.

Together, H.264 CRF23 (primary) and SAM2's semantic bottleneck (secondary) make this threat model unusually resistant to pixel-constrained adversarial perturbations.

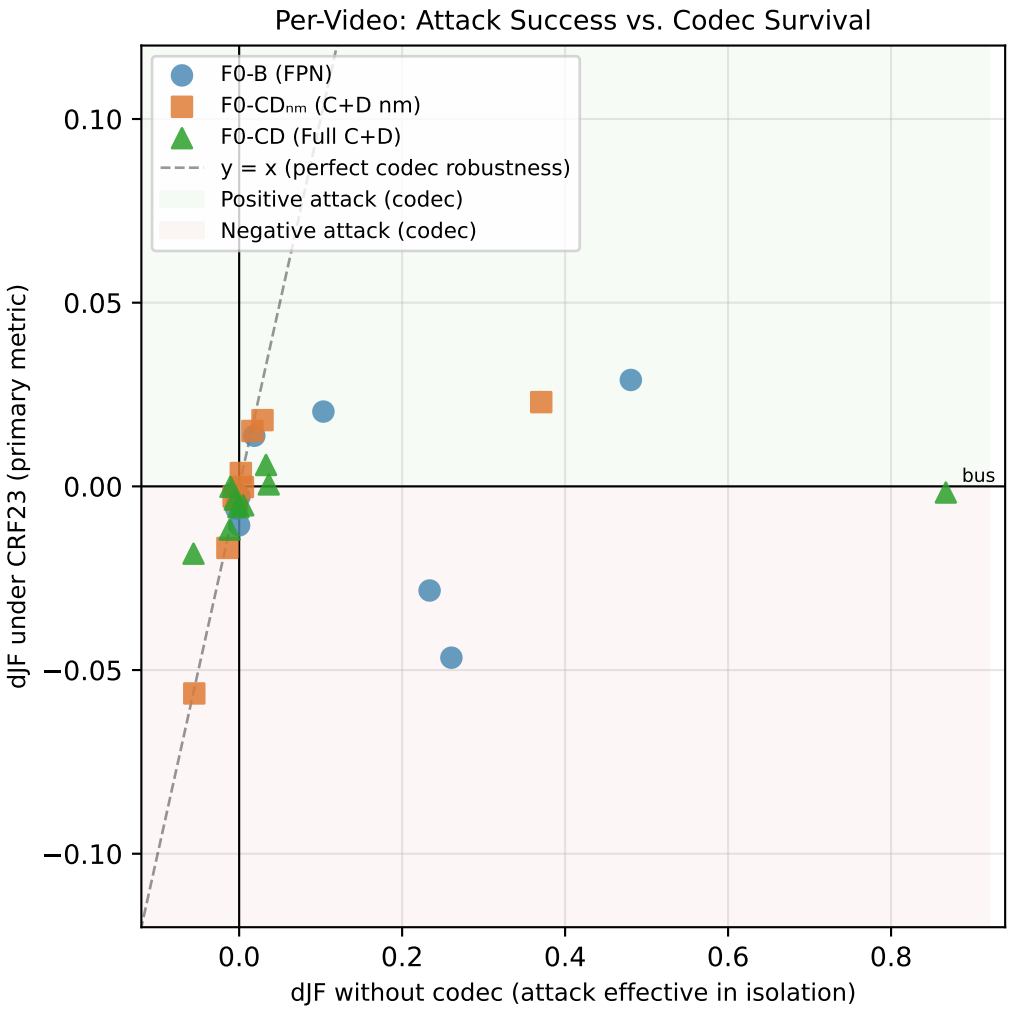

Figure 2: Per-video $\Delta\text{JF}_{\text{adv}}$ (x-axis) vs. $\Delta\text{JF}_{\text{auc}}$ (y-axis) for the three feature-space modes (27 data points). All attacks cluster near $y = 0$ regardless of attack mode or per-video attack strength. The bus data point ($x = 0.867$) illustrates that large optimizer-side attack strength does not predict codec survival.

8 Discussion, Limitations, and Implications

8.1 Implications for Privacy-Preserving Video

Our findings place a fundamental constraint on the adversarial approach to publisher-side privacy protection for SAM2-based tracking. Within the $\varepsilon = 8/255$ L_∞ budget, the standard video release pipeline (H.264 CRF23) acts as an effective adversarial purifier that the attacker cannot overcome with current gradient-based methods. Future work in this direction should consider:

- **Larger ε**: At sufficient perturbation magnitude, low-frequency pixel energy would survive codec. However, the SSIM and LPIPS constraints would also be violated, making the perturbation visible. The exact crossover point is an open question.
- **Semantic-level perturbations**: Perturbations that directly modify low-frequency video statistics (e.g., color distribution, spatial composition) may survive codec at lower ε, at the cost of changing video appearance more dramatically.
- **Alternative codecs**: VP9 and H.265 have different quantization profiles. CRF23 H.264 is the most relevant for practical video distribution, but robustness may differ across codec settings.

8.2 Implications for Adversarial Robustness Evaluation

Our results suggest that adversarial attacks against video segmentation and tracking should be evaluated under real codec conditions. UAP-SAM2 [Anonymous, 2025], DarkSAM [Zhang et al., 2024], and related work report results under standard (codec-free) conditions. If these systems are intended for deployment in real video pipelines, their attack effectiveness may be substantially reduced by the codec that is an unavoidable part of video distribution.

We recommend that video adversarial attack papers include a "post-codec" column in their main results tables, using real FFmpeg H.264 CRF23 (or the target deployment codec) rather than a differentiable proxy. Our differentiable proxy (YUV 4:2:0 + blur + noise + resize) is a reasonable training augmentation but is not a faithful evaluation substitute for real H.264.

8.3 Limitations

Single backbone. All experiments use SAM2.1 hiera_tiny. Larger models (SAM2-Small, SAM2-Large) have different feature-space characteristics and may exhibit different codec-robustness behavior.

Small validation set. Nine DAVIS videos provide clear trends but limited statistical power. Multi-dataset validation (YT-VOS, MOSE) would strengthen the negative result.

Surrogate gap in pixel-level attacks. The pixel-level training surrogate uses `logits.detach()` and a single rolling memory entry, approximating but not exactly replicating the full SAM2VideoPredictor computation graph. We cannot rule out that exact end-to-end gradient flow would produce different results, though the feature-space direct optimization experiments (which are closer to exact local optimization) also fail.

Dataset selection bias. The validity filter ($\mathrm{JF}_{\mathrm{clean}} \geq 0.5$) retains 9 DAVIS videos where SAM2 tracks well under clean conditions. Harder videos (lower $\mathrm{JF}_{\mathrm{clean}}$) might exhibit different codec-robustness dynamics; however, they also represent less relevant attack targets (if SAM2 already fails cleanly, perturbing them provides no privacy benefit).

Single CRF setting. Results are for CRF23. Very high quality (CRF18) would attenuate less; lower quality (CRF28) would attenuate more. A sweep across CRF values would strengthen the characterization of codec attenuation as a function of compression strength. CRF23 is the most practically relevant setting for public video distribution.

Adaptive codec-aware optimization (most important limitation). Our differentiable H.264 proxy is used as a *training regularizer*, not as a fully differentiable codec. A truly adaptive attacker with access to BPDA-style gradients [Athalye et al., 2018] through the real codec, or EOT-style robustness via compression augmentation, might produce perturbations with more low-frequency structure. We note that our feature-space direct optimization already uses the proxy during optimization, yet still fails; however, we cannot rule out that a more faithful gradient estimator through real H.264 would yield different results. This is the most important open question following from our work.

Scope. This is not an impossibility proof. We have shown that specific attack families fail within $\varepsilon = 8/255$. A fundamentally different attack paradigm—including stronger adaptive optimization, larger budgets, or semantic-level manipulation—might succeed.

8.4 Connection to Adversarial Purification

The codec-kill phenomenon we document is conceptually related to adversarial purification [Yoon et al., 2021, Nie et al., 2022], where a preprocessing step (typically a generative model) removes adversarial perturbations before the target classifier. Our finding shows this purification can occur "for free" as a side effect of lossy video compression. This has implications for the broader question of natural robustness: if a compression channel exists in the deployment pipeline, adversarial perturbations optimized for the pre-compression distribution may fail catastrophically in the post-compression regime.

9 Conclusion

We investigated whether pixel-constrained adversarial perturbations ($\varepsilon = 8/255$) can survive H.264 CRF23 to disrupt SAM2 video object tracking—a key requirement for publisher-side privacy protection. Through seven systematic experiments covering pixel-level CNN preprocessors and three feature-space optimization modes, we find a clear and consistent negative result within the tested attack families: H.264 CRF23 is an effective adversarial purifier that reduces attack effectiveness by 96–100% in all tested configurations.

The most striking evidence is the bus video case study: a single-frame perturbation that collapses SAM2 tracking from 93.5% to 6.8% JF score (without codec) produces zero practical effect after H.264 encoding. The optimizer is not the limiting factor—it finds effective perturbations. The codec destroys them.

We identify the root cause as the high-frequency concentration of L_∞-constrained perturbations, which are exactly the frequency components removed by DCT quantization in H.264 CRF23. Feature-space attacks targeting SAM2's lower-resolution memory representations do not escape this limitation, because the pixel-space gradient that drives them retains the same high-frequency structure.

Our results call for two changes in practice: (1) adversarial attack papers targeting video should include real-codec evaluation, and (2) adversarial privacy tool developers should treat codec robustness as a first-class design constraint rather than an afterthought. Within the current $\varepsilon = 8/255$ framework and the tested attack families, the codec appears to be an insurmountable barrier for gradient-based optimization without true end-to-end differentiability through the codec.

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